

# DISCRETE MATHEMATICS

## Chapter 1: The Foundations: Logic and Proofs

- **1.1: Propositional Logic**
  - 1 | 7 | 11 | 15 | 17 | 33 | 35
- **1.3: Propositional Equivalences**
  - 14 | 17 | 18 | 19 | 22 | 24 | 26 | 28
- **1.4: Predicates and Quantifiers**
  - 1 | 2 | 4 | 5 | 9 | 11 | 13 | 15 | 17 | 19 | 23 | 31 | 33
- **1.6: Rules of Inference**
  - 1 | 3 | 5 | 31
- **1.7: Introduction to Proofs**
  - 1 | 2 | 3 | 5 | 6 | 7 | 15 | 18a

## Chapter 2: Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

- **2.1: Sets**
  - 1 | 7 | 11 | 15 | 19 | 21 | 32 | 33 | 35 | 37
- **2.2: Set Operations**
  - 3 | 14 | 25 | 27 | 52 | 53 | 57
- **2.3: Functions**
  - 1 | 5 | 9 | 10 | 11 | 12 | 13 | 15 | 23 | 36
  - Reference book: 41 | 43
- **2.4: Sequences and Summations**
  - 1 | 2 | 3 | 9 | 10 | 13 | 15a | 15b | 15c | 17a | 17b | 17d | 29 | 31 | 33 | 39
- **2.5: Cardinality of Sets**
  - (Not specified)

## Chapter 5: Induction and Recursion

- **5.1: Mathematical Induction**
  - 3 | 15 | 18

## Chapter 6: Counting

- **6.1: The Basics of Counting**
  - 1 | 5 | 7 | 11 | 15 | 31
- **6.2: The Pigeonhole Principle**
  - 1 | 3 | 9

## Chapter 9: Relations

- **9.1: Relations and Their Properties**
  - 3 | 11
- **9.3: Representing Relations**
  - 1 | 2 | 3 | 7 | 19 | 21 | 23 | 25 | 31
- **9.5: Equivalence Relations**
  - 1

## Chapter 10: Graphs

- **10.1: Graphs and Graph Models**
  - 1 | 2 | 3 | 5 | 7 | 31 | 33 | 35

## Chapter 11: Trees

- (Details not provided)